

Macro-Conditional & Regime-Dependent Credit Risk Modeling

MScFE Capstone — Group 14074

- Taoufiq OUEDRAOGO
- Wickliff Siriga OGARI

$PD = Pr(\text{default} X_{\text{borrower}}, M_t, R_t)$	
Symbol	Meaning
X_{borrower}	Borrower-specific characteristics (FICO, DTI, income ...)
M_t	Macroeconomic variables at loan issuance time t
R_t	Latent market regime at time t (detected via HMM / GMM)

3-tier model hierarchy:

- Model 1 - Baseline** : $PD = f(X_{\text{borrower}})$
- Model 2 - Macro-Augmented** : $PD = f(X_{\text{borrower}}, M_t)$
- Model 3 - Regime-Dependent** : $PD = f(X_{\text{borrower}}, M_t, R_t)$

```
In [ ]: import warnings, os
warnings.filterwarnings("ignore")

import ssl
ssl._create_default_https_context = ssl._create_unverified_context

import numpy as np
import pandas as pd

import kagglehub

import seaborn as sns
import matplotlib.pyplot as plt

# --- ML / Stats ---
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_val_score
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    roc_auc_score, roc_curve, average_precision_score,
    precision_recall_curve, brier_score_loss,
    classification_report, confusion_matrix, ConfusionMatrixDisplay
)
from sklearn.calibration import calibration_curve, CalibratedClassifierCV
from sklearn.mixture import GaussianMixture
from sklearn.cluster import KMeans, AgglomerativeClustering

# --- XGBoost / LightGBM ---
#from xgboost import XGBClassifier
#import lightgbm as lgb

# --- HMM ---
#from hmmlearn import hmm

# --- Misc ---
from scipy import stats
from datetime import datetime
#import shap

# --- Reproducibility ---
SEED = 42
np.random.seed(SEED)

PALETTE = {"Base": "#487bd0", "Macro-Augmented": "#ee854a", "Regime-Dependent": "#6acc65"}
```

```
In [ ]:
```

1. Data Pipeline - Objective O1

Point-in-Time join ensures no look-ahead bias:

$$D_{merged} = \{(x_{i,t}, m_t) \mid \tau = \max(t_{macro} \leq t_{loan})\}$$

- Lending Club public dataset from [Kaggle](#)
- Macro data from [FRED API](#)

Load macro data

```
In [2]: def load_macro_data():
    """
    Load real macroeconomic data from FRED
    """
    import ssl
    ssl._create_default_https_context = ssl._create_unverified_context

    def get_fred_series(series_id):
        url = f"https://fred.stlouisfed.org/graph/fredgraph.csv?id={series_id}"
        df = pd.read_csv(url)
        df.columns = ["date", series_id]
        df["date"] = pd.to_datetime(df["date"])
        return df.set_index("date")

    # Retrieve key macro series
    fed = get_fred_series("DFF") # Fed Funds Rate
    unemp = get_fred_series("UNRATE") # Unemployment rate
    cpi = get_fred_series("CPIAUCSL") # Inflation index
    ind_prod = get_fred_series("INDPRO") # Industrial production
    gdp = get_fred_series("GDP") # GDP (quarterly)
    hy_spread = get_fred_series("BAMLH0A0HYM2") # High yield spread

    # Merge all
    macro = fed.join([unemp, cpi, ind_prod, gdp, hy_spread], how="outer")

    # Rename columns for consistency
    macro.columns = [
        "fed_funds",
        "unemp_rate",
        "cpi",
        "ind_prod",
        "real_gdp",
        "hy_spread"
    ]

    # Sort and clean
    macro = macro.sort_index()

    # Forward-fill missing values (critical for macro data)
    macro = macro.ffill()
    return macro.reset_index()
```

```
In [3]: macro_df = load_macro_data()
macro_df.head()
```

```
Out [3]:
```

	date	fed_funds	unemp_rate	cpi	ind_prod	real_gdp	hy_spread
0	1919-01-01	NaN	NaN	NaN	4.8739	NaN	NaN
1	1919-02-01	NaN	NaN	NaN	4.6585	NaN	NaN
2	1919-03-01	NaN	NaN	NaN	4.5238	NaN	NaN
3	1919-04-01	NaN	NaN	NaN	4.6046	NaN	NaN
4	1919-05-01	NaN	NaN	NaN	4.6315	NaN	NaN

Load loan data

```
In [4]: # Download latest version from kaggle
path = kagglehub.dataset_download("adarshsng/lending-club-loan-data-csv")

Warning: Looks like you're using an outdated `kagglehub` version (installed: 0.3.6), please consider upgrading to the latest version (1.0.0).
```

```
In [5]: print("Path to dataset files:", path)
Path to dataset files: /Users/taoufiq/.cache/kagglehub/datasets/adarshsng/lending-club-loan-data-csv/versions/1
```

```
In [6]: print(os.listdir(path))
['loan.csv', 'LCDataDictionary.xlsx']
```

```
In [37]: KEEP_COLS = [
    "issue_d", "loan_amnt", "term", "int_rate", "grade", "sub_grade",
    "emp_length", "home_ownership", "annual_inc", "dti", "delinq_2yrs",
    "#fico_range_low", "fico_range_high",
    "inq_last_6mths",
    "mths_since_last_delinq", "pub_rec_bankruptcies", "num_accts_ever_120_pd",
    "pct_tl_nvr_dlq", "loan_status", "earliest_cr_line",
]

file_path = os.path.join(path, "loan.csv")
loan_df = pd.read_csv(file_path, low_memory=False, nrows=1000) [KEEP_COLS]
print(loan_df.shape)
loan_df.head()
```

```
Out [37]:
```

	issue_d	loan_amnt	term	int_rate	grade	sub_grade	emp_length	home_ownership	annual_inc	dti	delinq_2yrs	inq_last_6mths	mths_since_last_delinq	pub_rec_bankruptcies	num_accts_ever
0	Dec-2018	2500	36 months	13.56	C	C1	10+ years	RENT	55000	18.24	0	1	NaN	1	
1	Dec-2018	30000	60 months	18.94	D	D2	10+ years	MORTGAGE	90000	26.52	0	0	71.0	1	
2	Dec-2018	5000	36 months	17.97	D	D1	6 years	MORTGAGE	59280	10.51	0	0	NaN	0	
3	Dec-2018	4000	36 months	18.94	D	D2	10+ years	MORTGAGE	92000	16.74	0	0	NaN	0	
4	Dec-2018	30000	60 months	16.14	C	C4	10+ years	MORTGAGE	57250	26.35	0	0	NaN	0	

```
In [35]: # Basic cleaning
loan_df["issue_d"] = pd.to_datetime(loan_df["issue_d"], errors='coerce')
loan_df = loan_df.sort_values("issue_d").reset_index(drop=True)

loan_df["term"] = loan_df["term"].str.extract("(\\d+)").astype(int)
loan_df["emp_length"] = loan_df["emp_length"].str.extract("(\\d+)").astype(float).fillna(0)
loan_df["earliest_credit_line"] = pd.to_datetime(loan_df["earliest_cr_line"], format="%b-%Y").dt.year
loan_df.head()
```

```
Out [35]:
```

	issue_d	loan_amnt	term	int_rate	grade	sub_grade	emp_length	home_ownership	annual_inc	dti	delinq_2yrs	inq_last_6mths	mths_since_last_delinq	pub_rec_bankruptcies	num_accts_ever
0	2018-12-01	2500	36	13.56	C	C1	10.0	RENT	55000	18.24	0	1	NaN	1	
1	2018-12-01	10000	36	6.46	A	A1	10.0	MORTGAGE	115000	10.41	2	1	18.0	0	
2	2018-12-01	30000	36	16.14	C	C4	1.0	MORTGAGE	90000	7.60	0	0	NaN	0	
3	2018-12-01	13000	36	8.19	A	A4	10.0	MORTGAGE	35000	22.57	0	0	69.0	0	
4	2018-12-01	24000	60	10.72	B	B2	10.0	MORTGAGE	76500	18.51	0	0	NaN	0	

```
In [ ]: # Define target
loan_df = loan_df[loan_df["loan_status"].isin(["Fully Paid", "Charged Off"])]
loan_df["loan_default"] = (loan_df["loan_status"] != "Charged Off").astype(int)

loan_df.head()
```

```
Out [ ]:
```

	issue_d	loan_amnt	term	int_rate	grade	sub_grade	emp_length	home_ownership	annual_inc	dti	delinq_2yrs	inq_last_6mths	mths_since_last_delinq	pub_rec_bankruptcies	num_accts_ever
51	2018-12-01	10450	60	12.98	B	B5	10.0	MORTGAGE	58000	21.17	0	1	NaN	0	
165	2018-12-01	5000	36	7.56	A	A3	5.0	MORTGAGE	98000	14.78	0	0	59.0	0	
231	2018-12-01	12950	36	7.56	A	A3	10.0	MORTGAGE	55000	19.88	0	1	NaN	0	
273	2018-12-01	10000	36	11.80	B	B4	1.0	RENT	100000	10.60	1	1	6.0	0	
375	2018-12-01	3500	36	18.94	D	D2	10.0	MORTGAGE	127300	17.91	2	2	3.0	0	

```
In [25]: def pit_join(loan_df, macro_df):
    """Merge macro onto loans using the last available macro observation
    at or before each loan's issue date - prevents data leakage."""
    merged = pd.merge_asof(
        loan_df.sort_values("issue_d"),
        macro_df.sort_values("date"),
        left_on="issue_d", right_on="date",
        direction="backward"
    ).drop(columns="date")
    return merged

data = pit_join(loan_df, macro_df)
data.head()
```

```
Out [25]:
```

	issue_d	loan_amnt	term	int_rate	grade	sub_grade	emp_length	home_ownership	annual_inc	dti	...	pct_tl_nvr_dlq	loan_status	earliest_cr_line	earliest_credit_line	fed_funds	unemp_rate
0	2018-12-01	2500	36	13.56	C	C1	10.0	RENT	55000	18.24	...	100.0	Current	Apr-2001	2001	2.2	3.9
1	2018-12-01	30000	36	16.14	C	C4	1.0	MORTGAGE	90000	7.60	...	100.0	Current	Apr-1991	1991	2.2	3.9
2	2018-12-01	13000	36	8.19	A	A4	10.0	MORTGAGE	35000	22.57	...	96.3	Current	Feb-2004	2004	2.2	3.9
3	2018-12-01	24000	60	10.72	B	B2	10.0	MORTGAGE	76500	18.51	...	100.0	Current	Jul-2005	2005	2.2	3.9
4	2018-12-01	6000	36	7.02	A	A2	10.0	MORTGAGE	20000	57.14	...	95.2	Current	Jul-2006	2006	2.2	3.9

5 rows x 25 columns

```
In [27]: def engineer_features(df):
    """Adds derived features used by both borrower and macro models."""
    df = df.copy()

    # Borrower
    df["credit_age"] = df["issue_d"].dt.year - df["earliest_credit_line"]
    df["high_dti"] = (df["dti"] > 30).astype(int)

    # Macro deltas (month-over-month changes - computed on sorted data)
    for col in ["fed_funds", "unemp_rate", "hy_spread", "cpi"]:
        df[f"{col}_delta"] = df[col].diff().fillna(0)

    # Macro composite stress index
    df["macro_stress"] = (
        df["unemp_rate"] / df["unemp_rate"].max() * 0.4 +
        df["hy_spread"] / df["hy_spread"].max() * 0.4 +
        (1 - df["fed_funds"] / df["fed_funds"].max()) * 0.2
    )

    return df

data = engineer_features(data)
data.head()
```

```
Out [27]:
```

	issue_d	loan_amnt	term	int_rate	grade	sub_grade	emp_length	home_ownership	annual_inc	dti	...	ind_prod	real_gdp	hy_spread	credit_age	high_dti	fed_funds_delta	unemp_rate_delta	hy_sprea
0	2018-12-01	2500	36	13.56	C	C1	10.0	RENT	55000	18.24	...	20917.867	NaN	17	0	0.0	0.0	0.0	
1	2018-12-01	30000	36	16.14	C	C4	1.0	MORTGAGE	90000	7.60	...	20917.867	NaN	27	0	0.0	0.0	0.0	
2	2018-12-01	13000	36	8.19	A	A4	10.0	MORTGAGE	35000	22.57	...	20917.867	NaN	14	0	0.0	0.0	0.0	
3	2018-12-01	24000	60	10.72	B	B2	10.0	MORTGAGE	76500	18.51	...	20917.867	NaN	13	0	0.0	0.0	0.0	
4	2018-12-01	6000	36	7.02	A	A2	10.0	MORTGAGE	20000	57.14	...	20917.867	NaN	12	1	0.0	0.0	0.0	

5 rows x 32 columns

```
In [29]: def build_target(df, seed=SEED):
    """Synthetic default label driven by both micro & macro signals."""
    np.random.seed(seed)
    logit = (
        # (720 - df["fico"]) / 70
        + (df["unemp_rate"] - 5.5) / 2.5
        + (df["hy_spread"] - 4.5) / 3
        + (df["dti"] - 20) / 25
        - df["annual_inc"].pipe(np.log) / 12
    )
    prob = 1 / (1 + np.exp(-logit))
    df["default"] = (prob > np.random.rand(len(df))).astype(int)
    return df

data = build_target(data)
data.head()
```

```
Out [29]:
```

	issue_d	loan_amnt	term	int_rate	grade	sub_grade	emp_length	home_ownership	annual_inc	dti	...	real_gdp	hy_spread	credit_age	high_dti	fed_funds_delta	unemp_rate_delta	hy_sprea
0	2018-12-01	2500	36	13.56	C	C1	10.0	RENT	55000	18.24	...	20917.867	NaN	17	0	0.0	0.0	0.0
1	2018-12-01	30000	36	16.14	C	C4	1.0	MORTGAGE	90000	7.60	...	20917.867	NaN	27	0	0.0	0.0	0.0
2	2018-12-01	13000	36	8.19	A	A4	10.0	MORTGAGE	35000	22.57	...	20917.867	NaN	14	0	0.0	0.0	0.0
3	2018-12-01	24000	60	10.72	B	B2	10.0	MORTGAGE	76500	18.51	...	20917.867	NaN	13	0	0.0	0.0	0.0
4	2018-12-01	6000	36	7.02	A	A2	10.0	MORTGAGE	20000	57.14	...	20917.867	NaN	12	1	0.0	0.0	0.0

5 rows x 33 columns